

NYU|Tandon

MA 2034

WORKSHEET 1

FALL 2026

Differential Equations Worksheet: Exponential Models**Total: 100 points****Name:** _____**Date:** _____

Instructions: Show all work and explain your reasoning when requested. **You must upload your work on this template and assign your work to the appropriate questions.**

Question 1. Alligator Population — Exponential Growth (40 points)

A protected population of alligators is modeled by

$$\frac{dP}{dt} = kP,$$

where $P(t)$ is the number of alligators t years after conservation efforts begin. Initially there are 400 alligators. After 5 years, the population has increased to 500.

(a) Interpretation of the Model**(5 points)**

Circle ALL true statements that apply to the differential equation

$$\frac{dP}{dt} = kP.$$

- A. The alligator population increases by exactly k alligators each year.
- B. The rate at which the alligator population changes is proportional to the current population.
- C. The alligator population will eventually approach k alligators.
- D. The percentage growth of the population increases at a constant rate.
- E. The alligator population increases only when $P = k$.

Choice: _____**Briefly explain your choice(s).**

(b) Solve the Differential Equation**(10 points)**

Starting with

$$\frac{dP}{dt} = kP,$$

solve the differential equation by separation of variables. Then use the initial condition

$$P(0) = 400$$

to obtain a solution containing only the unknown parameter k .

Show each major step.

Question 1 continued**(40 points total)****(c) Determine the Growth Constant****(5 points)**

Use the additional information

$$P(5) = 500$$

to determine k . Write the complete particular solution $P(t)$. Show all your work.

(d) Predict the Population**(5 points)**

According to this model, approximately how many alligators will there be after 15 years? Show all your work.

(e) Changing the Parameter k **(10 points)**

Suppose improved habitat causes the value of k to **double**, while the initial population remains 400.

1. Describe how doubling k affects the graph of $P(t)$.
2. Would doubling k cause the population to be twice as large at every future time? Explain mathematically or conceptually.

(f) Long-Term Behavior**(5 points)**

Determine

$$\lim_{t \rightarrow \infty} P(t).$$

Explain why this long-term prediction may be unrealistic for an actual alligator population.

Question 2. A Message from Space — Exponential Decay (30 points)

A distant spacecraft sends a continuous radio signal toward Earth. As its power source deteriorates, the strength $S(t)$ of the transmitted signal is modeled by

$$\frac{dS}{dt} = -kS, \quad k > 0.$$

Initially the signal strength is 80 units. Six years later it has fallen to 50 units.

(a) Find the Model (10 points)

Find k and determine the particular solution $S(t)$. Show enough work to justify your answer.

(b) Signal Strength of 10 Units (5 points)

According to the model, when will the signal strength fall to 10 units? Show all your work.

(c) Changing the Decay Constant**(10 points)**

Engineers redesign the power system so that the decay constant k is reduced by 50%.

1. Describe how $S(t)$ changes.
2. What happens to the time required for the signal to fall to one-half of its initial strength? Explain.

(d) Long-Term Behavior**(5 points)**

Determine

$$\lim_{t \rightarrow \infty} S(t).$$

Does the mathematical model predict that the signal ever becomes exactly zero at a finite time? Explain.

Question 3. A Meteorite in the Laboratory — Newton's Law of Heating & Cooling (30 points)

A meteorite recovered from Antarctica has been stored outside at -20°C . It is brought into a laboratory maintained at 20°C .

Its temperature is modeled by Newton's law of heating:

$$\frac{dT}{dt} = k(20 - T), \quad k > 0.$$

When the meteorite enters the laboratory,

$$T(0) = -20^{\circ}\text{C}.$$

After 30 minutes,

$$T(30) = 0^{\circ}\text{C}.$$

(a) Find the Model (10 points)

Determine k and find the particular solution $T(t)$. Show all your work.

(b) Reach 15°C (5 points)

According to the model, how long will it take the meteorite to reach 15°C ? Show all your work.

(c) Increasing k **(5 points)**

Suppose a fan is pointed at the meteorite, causing k to increase while the laboratory remains at 20°C .

Describe how increasing k affects the graph of $T(t)$.

Does increasing k change the eventual temperature of the meteorite? Explain.

(d) Changing the Room Temperature**(4 points)**

Now suppose the laboratory temperature is increased from 20°C to 25°C .

Without solving the new differential equation, explain what happens to the long-term behavior of the solution and determine

$$\lim_{t \rightarrow \infty} T(t).$$

(e) Sketch Solutions**(6 points)**

Include a sketch of a possible solution, $T(t)$, for each of the curves in parts (a), (c) and (d).